FileReader class to read the file from diskTh

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The **'FileReader'** class in Java is used for reading the contents of a file. It is a subclass of **'InputStreamReader'**. And as designed to read character files. The **'FileReader'** class reads the character from a file in the default character encoding.

Key features of 'FileReader' class

- **Reading character files: "FileReader"** is used to read the content of a file as characters.
- **Default encoding:** It uses the platform's default character encoding.
- **Subclass of 'InputStreamReader':** Inherits the method from **'InputStreamReader'**, which in turn is a subclass of **'Reader'**.

Commonly used methods

- **'int read():** Reads a single character and returns it as an integer. Returns '-1' at the end of the file.
- **'int read(char[] buf, int offset, int length)':** Reads character into a portion of an array.
- **'void close()':** Closest stream and releases any stream resource associated with it.

```
package FileHandling1.src;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.IOException;
import java.util.Scanner;
public class FileReaderOne {
    public static void main(String[] args) {
        FileReader fr = null;
        String fileName;
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the file name: ");
        fileName = sc.nextLine();
        try
        {
            fr = new FileReader("./DataFiles/"+fileName);
            int i;
            System.out.println("The content is: \n");
            while( (i = fr.read()) != -1 ){
                System.out.print((char)i);
            }
            fr.close();
        }
        catch(FileNotFoundException e){}
```

```
            catch(IOException e){}
            sc.close();
        }
    }
```

Output:

```
Enter the file name:
text.txt
The content is:

My name is Satyender
I am teaching Subhalakshmi
PS D:\Students\Students\SubhalakshmiUEM\Coding\JavaSem5>
owCodeDetailsInExceptionMessages' '-cp' 'C:\Users\bluej\Ap
aa70b1db4158\redhat.java\jdt_ws\JavaSem5_a8b33528\bin' 'Fi
Enter the file name:
test.txt
The content is:

My name is Satyender
Hello World!
```

Reading characters in char[]

```
package FileHandling1.src;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.IOException;
import java.util.Scanner;
public class FileReaderCharArray {
    public static void main(String[] args) {
        FileReader fr = null;
        String fileName;
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the file name: ");
        fileName = sc.nextLine();
        try {
            fr = new FileReader("./DataFiles/" + fileName);
            int i;
            char buf[] = new char[100] ;
            System.out.println("The content is: \n");
            while ((i = fr.read(buf)) != -1) {
                System.out.println("Number of charcter in this line: "+i+"\n");
                System.out.print(new String(buf, 0, i));
```

```
        }
        fr.close();
    } catch (FileNotFoundException e) {
    } catch (IOException e) {
    }
    sc.close();
  }
}
```

Output:

```
Enter the file name:
test.txt
The content is:

Number of charcter in this line: 100

My name is Satyender
Hello World!
Creates a new character-stream reader whose critical sections willNumber of charcter in this line: 100

 synchronize on the reader itself.
Creates a new character-stream reader whose critical sections wilNumber of charcter in this line: 36

l synchronize on the given object.
```

Summary

- The **'FileReader'** class is a convenient way to read character files.
- It reads file using the platforms default encoding.
- Provides method to read characters one by one or into an array.
- Properly handle exceptions to avoid resource leaks.

The 'FileReader' class is useful for reading simple text files in Java. For more complex needs such as reading files with specific encoding or working with buffered input, you might consider using **'InputStreamReader'** with **'BufferedReader'**.